

Mathematics — Session 4

Exponential & Logarithmic Functions — Full Worked Exams with Solutions

Exercise 1 — Multiple Choice Questions (3 points)

1	a and b are two positive real numbers. If $\ln(a) = 3$ and $\ln(b) = 4$, then $\ln(a^2/b) = \dots$	$\frac{9}{4}$	2	$\frac{3}{2}$
2	The number of solutions of the equation $\ln(x - 3) + \ln(x + 3) = 0$ is:	2	1	0
3	$f(x) = \ln(-\ln x)$. The domain of definition of f is:	$]0, +\infty[$	$]1, +\infty[$	$]0, 1[$

Solutions

1) Answer: (b) — $\ln(a^2/b) = 2$

$$\ln(a^2/b) = \ln(a^2) - \ln(b) = 2 \cdot \ln(a) - \ln(b) = 2(3) - 4 = 6 - 4 = 2.$$

2) Answer: (b) — 1 solution

$$\text{Domain: } x - 3 > 0 \text{ and } x + 3 > 0 \Rightarrow x > 3.$$

$$\ln(x - 3) + \ln(x + 3) = \ln[(x - 3)(x + 3)] = \ln(x^2 - 9) = 0$$

$$\Rightarrow x^2 - 9 = 1 \Rightarrow x^2 = 10 \Rightarrow x = \sqrt{10} \text{ or } x = -\sqrt{10}.$$

Only $x = \sqrt{10} \approx 3.16$ satisfies $x > 3$, so the equation has exactly one solution.

3) Answer: (c) — $]0, 1[$

We need $-\ln x > 0$ (so that $\ln(-\ln x)$ is defined) and $x > 0$ (domain of $\ln x$). $-\ln x > 0 \Leftrightarrow \ln x < 0 \Leftrightarrow 0 < x < 1$. Combined with $x > 0$, the domain is $]0, 1[$.

Exercise 2 — Study of $f(x) = e^{2x}/(e^{2x} - 1)$ (10 points)

Consider the function f defined over $]-\infty, 0[\cup]0, +\infty[$ by

$$f(x) = \frac{e^{2x}}{e^{2x} - 1}.$$

Let (C) be its representative curve

1) $\lim_{x \rightarrow 0^+} f(x)$ as $x \rightarrow 0^+$. Deduce an asymptote to (C) .

As $x \rightarrow 0^+$: numerator $e^0 = 1$, denominator $e^{2x} - 1 \rightarrow 0^+$. So $\frac{1}{0^+} = +\infty$.

Since $\lim_{x \rightarrow 0^+} f(x) = +\infty$ as $x \rightarrow 0^+$, the line $x = 0$ (the y -axis) is a vertical asymptote to (C) .

2) Show that $\lim_{x \rightarrow +\infty} f(x) = 1$ as $x \rightarrow +\infty$. Deduce an asymptote (d) to (C) .

Factor e^{2x} out of the numerator and denominator:

$$\frac{e^{2x}}{e^{2x} - 1} = \frac{e^{2x}(1)}{e^{2x}\left(1 - \frac{1}{e^{2x}}\right)} = \frac{1}{1 - \frac{1}{e^{2x}}}$$

As $x \rightarrow +\infty$, $e^x \rightarrow +\infty$ so $1/e^x \rightarrow 0$, therefore the limit $= \frac{1}{1 - 0} = 1$.

Since $\lim_{x \rightarrow +\infty} f(x) = 1$ as $x \rightarrow +\infty$, the line $(d): y = 1$ is a horizontal asymptote to (C) at $+\infty$.

3) a) Verify that $f(x) - 1 = 1/(e^{2x} - 1)$.

$$f(x) - 1 = \frac{e^{2x}}{e^{2x} - 1} - \frac{e^{2x} - 1}{e^{2x} - 1} = \frac{e^{2x} - e^{2x} + 1}{e^{2x} - 1} = \frac{1}{e^{2x} - 1} \quad \checkmark$$

b) Show that if $x > 0$ then $e^{2x} > 1$.

The function $x \mapsto e^{2x}$ is strictly increasing on $\mathbb{R}$, and $e^0 = 1$ (at $x = 0$). So for $x > 0$, since e^{2x} is strictly increasing, $e^{2x} > e^0 = 1$.

c) Deduce the position of (C) with respect to (d) over $]0, +\infty[$.

For $x > 0$, $e^{2x} - 1 > 0$ (from part b), so $f(x) - 1 = 1/(e^{2x} - 1) > 0$, i.e. $f(x) > 1$. Therefore (C) is located above the line $(d): y = 1$ on the interval $]0, +\infty[$.

4) a) Show that $f'(x) = -2e^{2x}/(e^{2x} - 1)^2$.

With $u = e^{2x}$, $v = e^{2x} - 1$, $u' = 2e^{2x}$, $v' = 2e^{2x}$, and $f = u/v$ so $f' = (u'v - v'u)/v^2$:

$$\begin{aligned} & \frac{2e^{2x}(e^{2x} - 1) - 2e^{2x} \cdot e^{2x}}{(e^{2x} - 1)^2} \\ &= \frac{2e^{4x} - 2e^{2x} - 2e^{4x}}{(e^{2x} - 1)^2} = \frac{-2e^{2x}}{(e^{2x} - 1)^2} \quad \checkmark \end{aligned}$$

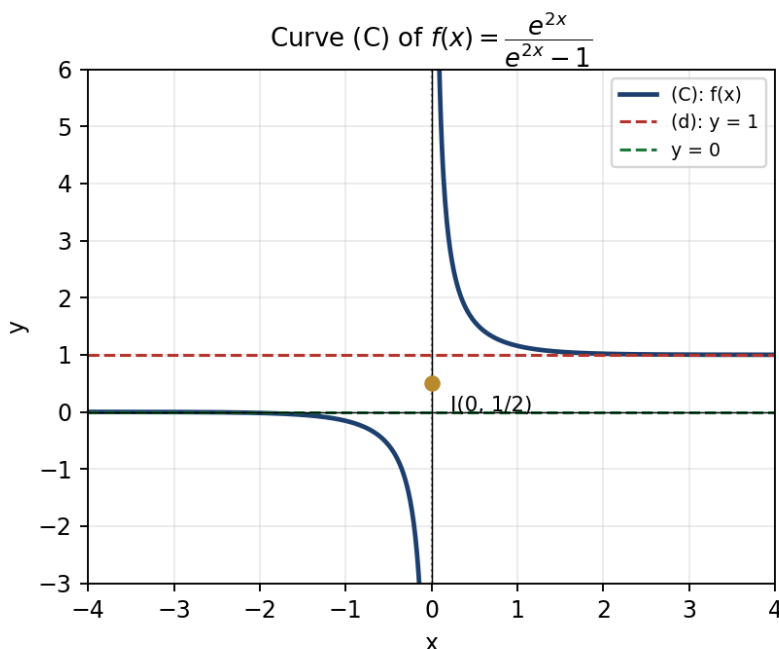
b) Table of variations of f

Since $e^{2x} > 0$ and $(e^{2x} - 1)^2 > 0$ for all $x \neq 0$, the numerator $-2e^{2x}$ is always negative, so $f'(x) < 0$ on both $]-\infty, 0[$ and $]0, +\infty[$: f is strictly decreasing on each interval.

Additional limit needed for the left branch: as $x \rightarrow -\infty$, $e^{2x} \rightarrow 0$, so $f(x) \rightarrow 0/(0-1) = 0$; and as $x \rightarrow 0^-$, $f(x) \rightarrow 1/0^- = -\infty$.

x	$-\infty$	0		$+\infty$
$f'(x)$	-		-	
$f(x)$	$0 \searrow -\infty$		$+\infty \searrow 1$	

5) Graph of (C) and (d)



(C): $f(x) = e^{2x}/(e^{2x} - 1)$, asymptotes $x = 0$, $y = 1$ and $y = 0$, center of symmetry $I(0, 1/2)$.

6) a) Show that $I(0, 1/2)$ is a center of symmetry of (C).

A point $I(a, b)$ is a center of symmetry of (C) when $f(2a - x) + f(x) = 2b$ for all x in the domain. Here $a = 0$, $b = 1/2$, so we must show $f(-x) + f(x) = 1$.

$$f(-x) = \frac{e^{-2x}}{e^{-2x} - 1} \quad (\text{multiply top and bottom by } e^{2x}) = \frac{1}{1 - e^{2x}} = \frac{-1}{e^{2x} - 1}$$

$$f(x) + f(-x) = \frac{e^{2x}}{e^{2x} - 1} + \frac{-1}{e^{2x} - 1} = \frac{e^{2x} - 1}{e^{2x} - 1} = 1$$

Since $f(-x) + f(x) = 1$ for all x , $I(0, 1/2)$ is indeed a center of symmetry of (C). ✓

b) Deduce an asymptote to (C) at $-\infty$.

By the point symmetry about $I(0, 1/2)$, the horizontal asymptote $y = 1$ at $+\infty$ corresponds to the horizontal asymptote $y = 2(1/2) - 1 = 0$ at $-\infty$. This matches the direct computation: $\lim_{x \rightarrow -\infty} f(x) = 0$ as $x \rightarrow -\infty$. So $y = 0$ is an asymptote to (C) at $-\infty$.

Exercise 3 — Multiple Choice Questions (4 points)

N°	Question	a	b	c
1	The domain of definition of the function f given by $f(x) = \frac{x}{1 - \ln x}$ is:	$[0, +\infty[$	$[e, +\infty[$	$]0, e[\cup]e, +\infty[$
2	The derivative of f defined over $]1/2, +\infty[$ as $f(x) = \ln(2x-1) - \ln(2x+1)$ is:	$\frac{4}{4x^2 - 1}$	$\frac{2}{4x^2 - 1}$	$\frac{2}{\ln(2x-1)} - \frac{2}{\ln(2x+1)}$
3	The solution of the equation $\ln(x + e) = 1$ is:	0	1	$1 - e$
4	The set of solutions of the inequality $e^{x+4} > 1$ is:	$] -4, +\infty[$	$] -\infty, -4[$	$[-4, +\infty[$

Solutions

1) Answer: (c) — $]0, e[\cup]e, +\infty[$

We need $x > 0$ (domain of $\ln x$) and $1 - \ln x \neq 0$, i.e. $\ln x \neq 1$, i.e. $x \neq e$. Combining both conditions gives the domain $]0, e[\cup]e, +\infty[$.

2) Answer: (a) — $\frac{4}{4x^2-1}$

$$f'(x) = \frac{2}{2x-1} - \frac{2}{2x+1} = \frac{2(2x+1) - 2(2x-1)}{(2x-1)(2x+1)} = \frac{4x+2-4x+2}{4x^2-1} = \frac{4}{4x^2-1}$$

3) Answer: (a) — $x = 0$

$$\ln(x + e) = 1 \Rightarrow x + e = e^1 = e \Rightarrow x = e - e = 0.$$

4) Answer: (a) — $] -4, +\infty[$

$$e^{x+4} > 1 = e^0 \Rightarrow x + 4 > 0 \Rightarrow x > -4. \text{ So } S =] -4, +\infty[.$$

Exercise 4 — Functions $g(x)$ and $f(x) = x + 2 + xe^{-x+1}$ (10 points)

Part A: $g(x) = 1 + (-x + 1)e^{-x+1}$ (functions study)

Consider g defined over $]-\infty, +\infty[$ by $g(x) = 1 + (-x + 1)e^{-x+1}$. Given $g'(x) = (x - 2)e^{-x+1}$.

1) Table of variations of g

Since $e^{-x+1} > 0$ always, the sign of $g'(x)$ is the sign of $(x - 2)$: negative for $x < 2$, zero at $x = 2$, positive for $x > 2$. So g is decreasing on $]-\infty, 2[$ and increasing on $]2, +\infty[$, with a minimum at $x = 2$:

$$g(2) = 1 + (-2+1)e^{-2+1} = 1 - e^{-1} \approx 0.63.$$

x	$-\infty$	2	$+\infty$
$g'(x)$	$-$	0	$+$
$g(x)$	$\searrow g(2)=1-e^{-1}$		$\nearrow$

2) Verify that $g(x) \geq 1 - e^{-1}$ for all real x .

From the table above, $g(2) = 1 - e^{-1}$ is the global minimum value of g on $\mathbb{R}$ (g decreases then increases around $x = 2$). Therefore $g(x) \geq g(2) = 1 - e^{-1}$ for all real x .

Part B: $f(x) = x + 2 + xe^{-x+1}$

Consider f defined over $]-\infty, +\infty[$ by $f(x) = x + 2 + xe^{-x+1}$. Let (C) be its curve and (d) : $y = x + 2$.

1) Determine $\lim_{x \rightarrow -\infty} f(x)$ as $x \rightarrow -\infty$. Calculate $f(-1)$ and $f(1)$.

As $x \rightarrow -\infty$: $-x + 1 \rightarrow +\infty$ so $e^{-x+1} \rightarrow +\infty$, and $x \rightarrow -\infty$, so $x \cdot e^{-x+1} \rightarrow (-\infty) \cdot (+\infty) = -\infty$. Also $x + 2 \rightarrow -\infty$. So $\lim_{x \rightarrow -\infty} f(x) = -\infty$ as $x \rightarrow -\infty$.

$$f(-1) = -1 + 2 + (-1)e^{1+1} = 1 - e^2 \approx -6.38.$$

$$f(1) = 1 + 2 + (1)e^{-1+1} = 3 + e^0 = 3 + 1 = 4.$$

2) a) Show that $\lim_{x \rightarrow +\infty} f(x) = +\infty$ as $x \rightarrow +\infty$.

As $x \rightarrow +\infty$, $x + 2 \rightarrow +\infty$, and by the standard comparison limit $\lim_{x \rightarrow +\infty} (x \cdot e^{-x}) = 0$ as $x \rightarrow +\infty$ (exponential decay dominates), we get $x \cdot e^{-x+1} = e \cdot (x \cdot e^{-x}) \rightarrow 0$. Therefore $f(x) = (x + 2) + xe^{-x+1} \rightarrow +\infty + 0 = +\infty$.

b) Study, according to the values of x , the relative position of (C) with respect to (d) .

$f(x) - (x + 2) = xe^{-x+1}$. Since $e^{-x+1} > 0$ always, the sign of $f(x) - (x + 2)$ is the sign of x :

- For $x < 0$: $f(x) - (x + 2) < 0 \rightarrow (C)$ is below (d) .
- For $x = 0$: $f(x) = x + 2 \rightarrow (C)$ meets (d) (point of intersection).
- For $x > 0$: $f(x) - (x + 2) > 0 \rightarrow (C)$ is above (d) .

c) Show that (d) is an asymptote to (C) at $+\infty$.

$\lim_{x \rightarrow +\infty} [f(x) - (x + 2)] = \lim_{x \rightarrow +\infty} x \cdot e^{-x+1} = 0$ as $x \rightarrow +\infty$ (already shown in 2a). Since this difference tends to 0, the line (d) : $y = x + 2$ is an asymptote to (C) at $+\infty$.

3) a) Verify that $f'(x) = g(x)$. Deduce that f is strictly increasing over $\mathbb{R}$.

$f(x) = x + 2 + xe^{-x+1}$. Using the product rule on xe^{-x+1} ($u = x$, $v = e^{-x+1}$, $u' = 1$, $v' = -e^{-x+1}$):

$$f'(x) = 1 + (1)e^{-x+1} + (-e^{-x+1})(x) = 1 + e^{-x+1} - xe^{-x+1}$$

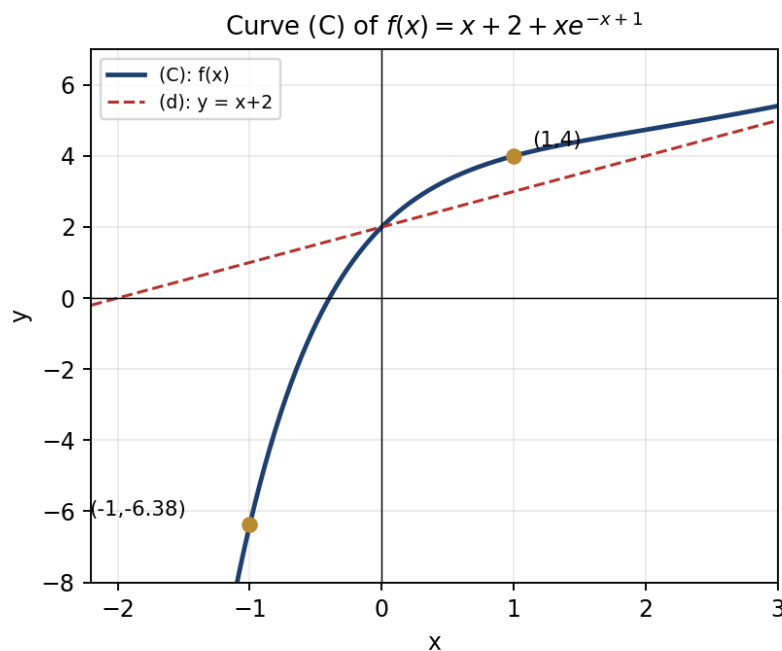
$$= 1 + (1-x)e^{-x+1} = 1 + (-x+1)e^{-x+1} = g(x). \quad \checkmark$$

Since $g(x) \geq 1 - e^{-1} \approx 0.63 > 0$ for all x (Part A, question 2), we get $f'(x) = g(x) > 0$ for all x , so f is strictly increasing on $]-\infty, +\infty[$.

b) Table of variations of f

x	$-\infty$	$+\infty$
$f'(x)$	$+$	
$f(x)$	$-\infty \nearrow +\infty$	

4) Graph of (d) and (C)



(C): $f(x) = x + 2 + xe^{-x+1}$, asymptote (d): $y = x + 2$ at $+\infty$, with $f(-1) \approx -6.38$ and $f(1) = 4$.

5) Line (D): $y = x + m$. Find m for which (D) is tangent to (C).

The tangent to (C) at the point of abscissa a has equation $y = f'(a)(x - a) + f(a)$. For this tangent to be parallel to (D) (both have slope 1), we need $f'(a) = 1$, i.e. $g(a) = 1$:

$$1 + e^{-a+1}(1 - a) = 1 \Rightarrow e^{-a+1}(1 - a) = 0.$$

Since $e^{-a+1} \neq 0$ for any a , we need $1 - a = 0$, i.e. $a = 1$.

The tangent at $a = 1$: $y = f'(1)(x - 1) + f(1) = 1 \cdot (x - 1) + 4 = x + 3$.

Comparing with $y = x + m$, we get $m = 3$.

6) k is defined over $]-\infty, +\infty[$ with the given table of variations (increasing to $k(3) = e^{-2}$, then decreasing to 0). Find the domain of $h(x) = 1/(k(x) - g(x))$.

We need $k(x) - g(x) \neq 0$, i.e. $k(x) \neq g(x)$.

- From Part A: $g(x) \geq 1 - e^{-1} \approx 0.63$ for all x .

- From the table of k : the maximum value of k is $k(3) = e^{-2} \approx 0.13$, so $k(x) \leq e^{-2}$ for all x .

Since $e^{-2} \approx 0.13 < 0.63 \approx 1 - e^{-1}$, we have $k(x) \leq e^{-2} < 1 - e^{-1} \leq g(x)$ for all x , so $k(x) < g(x)$ always, meaning $k(x) \neq g(x)$ for every real x .

Therefore $k(x) - g(x)$ never vanishes, and the domain of h is all of $\mathbb{R}$: $Dh =]-\infty, +\infty[$.